



SCI_LAUNCH_A2_W05 · LAUNCH Science

Autumn 2 Week 5 teacher note

Objective (verbatim): Complete a genetic cross (digital Punnett square).

Scheme of work: LAUNCH scheme of work, Autumn 2 week 5.

Pupil words: I am learning to calculate genetic probability with a Punnett square and evaluate its limits.

Reading gate: LAUNCH 11+; rough FK grade 5.1, age proxy ≈ 11.1 , 994 pupil-visible words excluding marked adult guidance. Formula is a rough screen, not an individual reading assessment.

Pearson foundation route: GCSE Biology 1BI0 Paper 1 and Combined Science 1SC0 Biology Paper 1, shared points 3.12–3.14 and 3.16 (shared Foundation). Official specification links: <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-biology-spec.pdf> and <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-combinedscience-spec.pdf>. The workbook's teaching approach is not a new specification point. No Biology-only B/HT point is assumed for Combined Foundation.

Gatsby 4: A genetic counsellor explains probabilities without claiming a family outcome is certain.

Sequence, success, retrieval

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40 minutes. Three one-minute resets are inside We do, Independent and Review; Arrival and resets make about eight minutes for settling and transitions.

Success at Starter, checked at Exit: I can pair one allele from each parent.; I can find the aa chance in four squares.; I can evaluate why probability does not guarantee an exact count.

Retrieval from SoW: "last term" uses the Autumn 1 block in the supplied 2026/27 workbook. If a literal previous academic term is required, staff must check actual teaching rather than invent it.

When	Question	Answer	Exact SoW cell
last lesson	What is a gene?	A section of DNA that codes for a protein.	LAUNCH scheme of work, Autumn 2 week 4
2-4 weeks	What do stem cells do in growth?	They divide and some become specialised.	LAUNCH scheme of work, Autumn 2 week 2
last term	Where are chromosomes found in an animal cell?	In the nucleus.	LAUNCH scheme of work, Autumn 1 week 1

Word meanings and examples (same in lesson and organiser):

Word	Tier	Definition	Example sentence
allele	3	a version of a gene	A and a represent two alleles in our invented model.
genotype	3	the pair of alleles shown by letters	Aa is one possible genotype in this model.
dominant	3	an allele shown by one copy in this simple model	A is dominant in our invented flower model.



Word	Tier	Definition	Example sentence
recessive	3	an allele shown when two copies are present in this model	aa shows the recessive version in our model.
Punnett square	3	a grid that shows possible allele pairs	The Punnett square for Aa × aa has four cells.

Faded example: I do full: Place A/a atop the grid for one parent; place a/a at its side for the other. Pair one allele from each edge: Aa, aa, Aa, aa. Two of four boxes are aa: a 1 in 2 chance per offspring, not an exact family count. We do part-done: Aa × aa: top A, a; side a, a. First row Aa, aa; second row _, _. Chance of aa: _ out of 4. Removed step: Combine one top allele and one side allele in each second-row square; count two aa squares. Independent uses no filled answer. Fade next: Next lesson hide the top A/a headings; pupil selects them from the parent genotype before filling the grid.

Post-I-do show-me: Show me: which allele can the aa parent pass on? Expected: Only a. Every pupil shows a board, point or card. Re-model the named step if insecure.

F/M/H and named stretch

We do F support: Place one letter tile from the top and one from the side into a single square. Point to the next pair.

We do M: Explain why the two aa squares represent half the model outcomes.

We do H stretch: Predict the aa chance for Aa × Aa before checking a second model. Explain the change.

Independent F: Complete Aa × aa with allele tiles. Stem: Top _ and side _ make _.

Independent M: Calculate and justify the aa probability from the grid. Stem: I count _ aa squares out of four, so ...

Independent H: Evaluate an exact-count claim about four real offspring. Stem: The grid shows a chance, not a promise, because ...

Go deeper: Compare aa chances for Aa × Aa and Aa × aa with a second grid.

Named early-finisher stretch — Faulty headings audit: Correct a faulty grid that gives an Aa parent two A headings; explain its wrong probability. This is a harder correction/evaluation, not a repeated item.

On-screen interactions: Complete the 2 × 2 digital grid. Choose the genotype in each square. uses native selects with a check/reset and aria-live feedback; We do 2 is a native 4-option all-pupil hinge without auto-grading. No storage. Original labelled diagram is simplified, not to scale. Its labels are checked in the fact ledger.

SoW practical status: no core practical is claimed. Prepared model pieces are an optional handling rehearsal with the pupil-facing Risk/Safe lines below.

AfL hinge and misconceptions

Question at We do 2: New cross: Aa × Aa. What is the chance of aa for each offspring?

Option	Correct?	Diagnosis
A. 1 in 4 for each offspring.	yes	Uses the new case accurately.
B. 1 in 2 for each offspring.	no	Counts the two heterozygous squares rather than the one aa square.



Option	Correct?	Diagnosis
C. 3 in 4 for each offspring.	no	Counts all outcomes except aa rather than the recessive genotype.
D. Certain for every single offspring.	no	Treats one possible aa square as a guaranteed outcome.

Decision: About 3 in 4 or more right → move on. Otherwise re-model the Aa × aa top and side headings with two colours, pair one allele in each square, then ask the parallel hinge. No automatic mark or saved response. Parallel hinge: Third cross: AA × aa. What is the chance of Aa for each offspring? Answer: certain; every possible pair contains one A and one a. This does not set an exact family size.

Misconception	How to spot	Correction
A possible aa outcome is certain for every offspring.	Pupil chooses D, making the aa outcome certain.	Each offspring has one chance in four in the model; it is not guaranteed.
The two Aa cells in a cross count as aa.	Pupil chooses B, counting Aa squares as aa.	Read the two allele symbols in each square; only one square is aa.

Planned adaptations

Broad area	EEF five-a-day	Concrete action
Cognition & Learning	scaffolding	Keep the four-cell allele model in view; remove one cue for We do, then ask pupils to repeat a fresh example without it.
Communication & Interaction	flexible grouping	Offer the key word card and a private partner turn before speaking to the group; accept point, sign or AAC.
SEMH	cognitive/metacognitive strategies	Check in before the top question and hinge; offer a private first answer, agreed pause and one-step return.
Sensory/Physical	technology	Provide an enlarged high-contrast model and keyboard focus; allow speech-to-text or exact scribing.
EAL (language, not SEN)	explicit instruction	Pre-teach defined science words with examples; let pupils plan in any language while keeping M/H reasoning open.

EAL language support remains available at M and H; the challenge stays open. Speak before write; the organiser's "My language" column is blank, without machine translation. Accept point, sign, speak, draw, type, write, exact scribing and staff-captured photo. Offer speech-to-text, TTS and immersive reader.

TA Focus and prompt ladder

At Independent choose at lesson start: teacher works with supported pupils and TA holds group, or TA supports and teacher holds group. Swap across the week. No public cold-calling is required; this is an invitation bank.

I do: "Watch one letter from each parent enter one square."

We do: "Which edge has not supplied a letter yet?"



Independent: "Point to one finished square. How will you use that step next?"

Least prompt first; no physical prompting: Wait five seconds for an independent tile move. → Point to the unfinished square and the two edges. → Ask "Which one allele can this parent pass on?" → Model one edge-to-square move; pupil repeats with another square. → Complete one square together, then pupil tries a fresh one alone.

Exit handoff: adult states deepest prompt level, valid response mode and what pupil did independently; other adult chooses the next prompt.

Route	Invitation	Expected answer
F	What can the aa parent pass on?	a only.
F	How many cells are in a two-by-two grid?	Four.
M	Why are two boxes aa?	The a from the Aa parent can pair with either a from the aa parent.
M	What chance does two out of four represent?	One in two for each offspring in this model.
H	Why is a probability not a fixed count in four offspring?	Each offspring is a separate chance; the grid gives possible proportions.
H	What changes with Aa × Aa?	The aa chance becomes one in four in the simple model.

Feedback, EHCP broad areas and regulation

Codes exactly: VF · WS · I · NS(+) · E · R · // · ?. VF at We do or Review; WS when supported with recorded prompt level; I when independent at Independent/Exit; NS(+) precise next step at Exit; // a real self-edit at Review; ? adult reads the changed words back on request; R after the pupil responds to NS(+) next lesson and staff judge, never automatic; E only for an actual curriculum observation.

What I record and where: note hinge response split and the named misconception on the staff quick-record grid; record actual code and prompt level in feedback. E: evidence captured on the school's evidence app. Never gather evidence during distress or store pupil data in HTML.

Pathways: high support, VF/WS regular. Build: WS/I mix. Grow: mostly independent, self-edit and response. Launch: independent self-assessment.

Regulation: Check in before the hinge, top rung and Independent. At the agreed pause signal give space and a single return step. Collect no evidence while a pupil is distressed. Re-enter at the last completed step, with one small choice and private correction.

lesson record poster: LAUNCH_SCI_A2_W05. Staff device or projector only; pupil phones not required.

Lundy Exit: pupil can say, point, sign, draw or write; teacher hears the answer and changes the next model/prompt. Four answers: goal at top, actual Review feedback, Exit NS(+), success criteria at Starter and Exit.

Sources, checks and limits

Every scientific and diagram label claim is listed in LESSON_SOURCE.json with a named official source and status. False distractors are deliberately incorrect and diagnosed. Read source claims in context; no medical claim about an individual. The A/a flower example is an invented single-gene teaching model. It says nothing about a real species or person.

Prepared model handling · practical safety

Kit: Large A and a letter tiles, prepared Punnett grid, tray



Method: Place one prepared tile from each parent into each grid square.

Pupil Risk: Small loose letter tiles could be mouthed or fall.

Pupil Safe: Use large tiles on a tray. Keep them out of mouths and collect them after.

Staff confirm the local school risk assessment and exact kit; this does not assert a SoW core practical.

SLT planning and evaluation

Lesson code: SCI-LAUNCH-A2-W05

SoW reference: LAUNCH scheme of work, Autumn 2 week 5

Exact SoW outcome: Complete a genetic cross (digital Punnett square).

Evidence (scheme of work): Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Evidence in this lesson: Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A2-W05. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Look at this week's letter tiles A and a. Point to one detail you notice. Staff: Offer a private, predictable start. Do not require previous knowledge.

Starter (3 min): Recall three ideas from earlier lessons. Show one answer at a time. Staff: All pupils respond; reveal the answers only after attempts. Revisit success criteria.

I Do (5 min): Watch the full model. Point to the step that answers today's question. Staff: Model one complete example; check every pupil's response to the show-me question.

We Do (7 min): Complete the missing step in the part-done model. Speak, then match the cards. Staff: Wait five seconds, gesture, then cue if needed. Support the route without supplying the missing answer.

We Do 2 (5 min): Choose one hinge answer. Show it at the same time as the group. Staff: Collect simultaneous responses, apply the decision rule and parallel hinge in the staff note.

Independent (8 min): Calculate the aa probability for $Aa \times aa$. Evaluate an exact-count claim about real offspring. Staff: At lesson start choose one adult with supported pupils and one holding the group; swap during the week. Record independence and prompt level.

Review (4 min): Tell a partner what you changed after feedback. Mark a real edit // and read it back when asked ?. Staff: Give a precise VF or NS(+) only when it actually happens. Invite a true self-edit and read-back.

Exit (3 min): Show what you know with a model and explain one reason. Staff: Hear the answer privately in a valid mode and use it to change the next example or prompt.

Planned adaptation — cognition learning: Keep the four-cell allele model in view; remove one cue for We do, then ask pupils to repeat a fresh example without it.

Planned adaptation — communication interaction: Offer the key word card and a private partner turn before speaking to the group; accept point, sign or AAC.

Planned adaptation — semh: Check in before the top question and hinge; offer a private first answer, agreed pause and one-step return.



Planned adaptation — sensory physical: Provide an enlarged high-contrast model and keyboard focus; allow speech-to-text or exact scribing.

Planned adaptation — eal: Pre-teach defined science words with examples; let pupils plan in any language while keeping M/H reasoning open.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: flexible grouping; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Calculate the aa probability for $Aa \times aa$. Evaluate an exact-count claim about real offspring. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W06. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W06: Research a genetic condition online; present.

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Combine one top allele and one side allele in each second-row square; count two aa squares. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "New cross: $Aa \times Aa$. What is the chance of aa for each offspring?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Look at this week's letter tiles A and a. Point to one detail you notice. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Calculate the aa probability for $Aa \times aa$. Evaluate an exact-count claim about real offspring. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use allele, genotype, dominant, recessive, Punnett square with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.